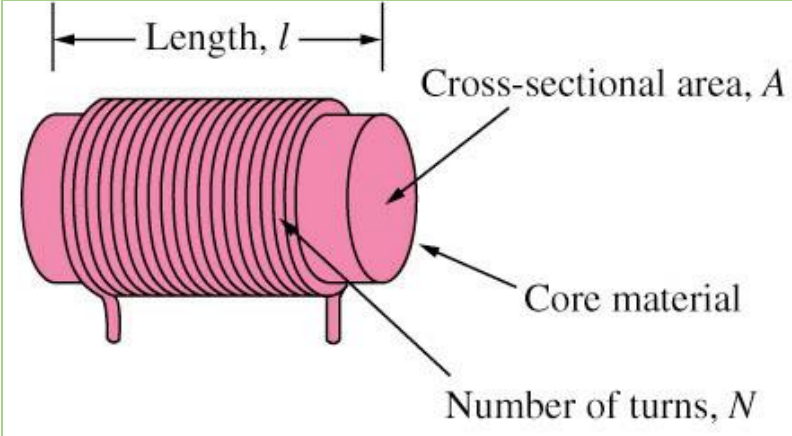


L#23: Inductors

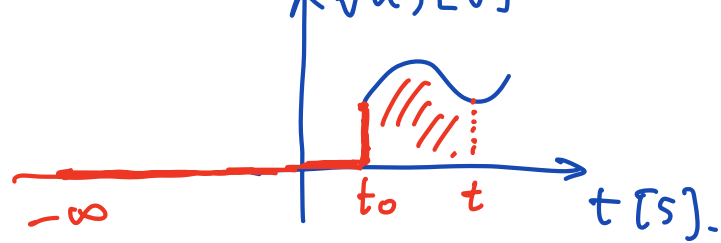
Physical characteristics

- An inductor is a passive element designed to store energy in its magnetic field.
- An inductor consists of a coil of conducting wire, with Core from iron, steel, plastic or air,

	$L = \frac{N^2 \mu A}{l} [H]$ $\frac{H}{m}$ $\uparrow$ m^2
N:	
μ: $\propto [H/m]$	
A:	
l:	
μ_r:	
μ_0: $4\pi \times 10^{-7} H/m$	

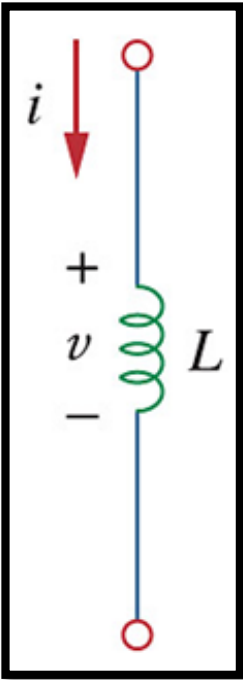
$$4\pi \times 10^{-7} H/m$$

$$\mu_r = \frac{\mu}{\mu_0}$$



Electrical characteristics

- Inductance (L): measured in Henry (H).



Voltage:

$$v(t) = L \frac{di(t)}{dt}$$

Current:

$$\int_{-\infty}^t v(\tau) d\tau = \int_{-\infty}^t L \frac{di(\tau)}{d\tau} d\tau$$

$$\int_{-\infty}^t v(\tau) d\tau = L i(t)$$

$$i(t) = \frac{1}{L} \int_{-\infty}^t v(\tau) d\tau = \frac{1}{L} \int_{-\infty}^{t_0} v(\tau) d\tau + \frac{1}{L} \int_{t_0}^t v(\tau) d\tau$$

$$i(t) = i(t_0) + \frac{1}{L} \int_{t_0}^t v(\tau) d\tau$$

$$\underline{i(t_0) = 0.}$$

Energy:

$$W = \int_{-\infty}^t L i(\tau) \cdot \frac{di(\tau)}{d\tau} d\tau$$

$$= \int_{i(-\infty)}^{i(t)} L i di$$

$$= \underline{\underline{\frac{1}{2} L i^2}}$$

$$\int x dx = \frac{1}{2} x^2$$

$$\left[\frac{1}{2} x^2 \right]' = \underline{\underline{x}}$$

Important characteristics of inductors:

- It is a short circuit for DC.

$$v(t) = L \frac{di(t)}{dt}, \text{ DC, } i(t) \rightarrow 0 \quad v(t) \rightarrow 0.$$

- Its current cannot change abruptly.

$$\frac{di(t)}{dt} \rightarrow \infty, \quad v(t) \rightarrow \infty.$$

- It does not dissipate energy but only store energy when charging or release energy when discharging.

Given $i(t)$ through a 3 H inductor is 4 A.

How much energy is consumed by L?
▲
stored

$$W = \frac{1}{2} L \cdot i^2 \quad [\text{J}].$$

Examples #1-2

#1: The current through a 1 mH inductor is $i(t) = 20 \cos 100t$ mA. Calculate the voltage and the stored energy.

$$\begin{aligned} v(t) &= L \frac{di}{dt} = 1 \times 10^{-3} [20 \cos 100t]' \cdot 10^{-3} \\ &= 10^{-3} \cdot 20(-\sin 100t) \cdot 100 \cdot 10^{-3} \\ &= -2 \sin 100t \text{ [mV]} \end{aligned}$$

$$W = \frac{1}{2} L i^2 = \frac{1}{2} \times 10^{-3} \times [20 \cos 100t \times 10^{-3}]^2 \text{ [J]}.$$

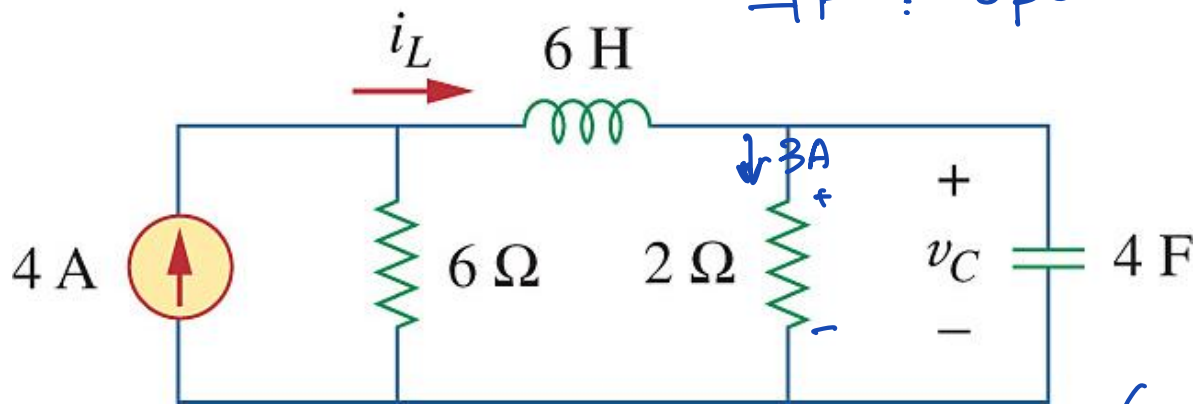
#2: The voltage across a 2 H inductor is $v(t) = 10(1 - t)$ V. Determine the current through the inductor at $t = 4$ s if $i(0) = 2$ A.

$$\begin{aligned} i(t) &= \frac{1}{L} \int_{t_0}^t v(\tau) d\tau + i(t_0), \quad t_0 = 0 \text{ [s]} \\ &= \frac{1}{2} \int_0^t 10(1 - \tau) d\tau + 2 \\ &= \frac{1}{2} \int_0^t (10 - 10\tau) d\tau + 2 \\ &= \frac{1}{2} [10\tau - 5\tau^2] \Big|_0^t + 2 \\ &= \frac{1}{2} [10t - 5t^2 - (10 \times 0 - 5 \times 0^2)] + 2 \\ &= \frac{1}{2} (10t - 5t^2) + 2 \end{aligned}$$

$$i(4) = \frac{1}{2} (10 \times 4 - 5 \times 4^2) + 2 = -18 \text{ A}.$$

Example #3: Determine v_C , i_L and the energy stored in both the capacitor and the inductor for the DC circuit below.

$\text{---} \text{---} \text{---}$: short
 $\text{---} | \text{---}$: open



$$W_L = \frac{1}{2} L i^2$$

$$W_C = \frac{1}{2} C v^2$$

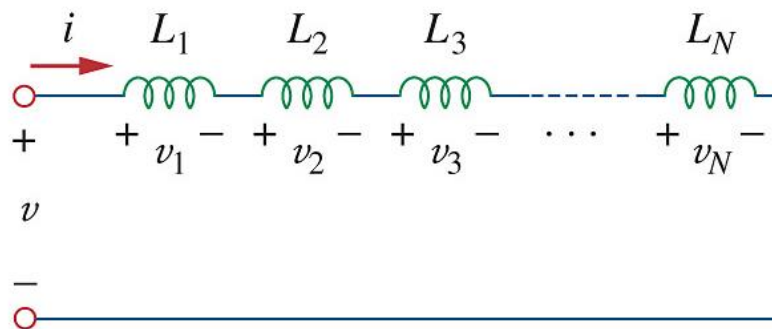
$$i_L = 4 \cdot \frac{6}{6+2} = 3 \text{ A}$$

$$v_C = 6 \text{ V}$$

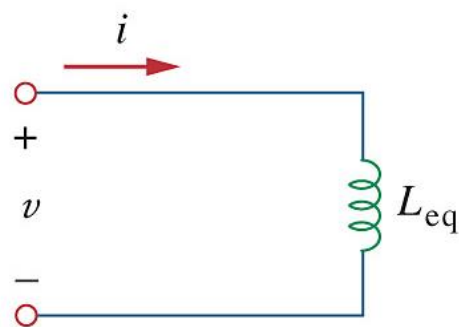
$$W_L = \frac{1}{2} \times 6 \times 3^2 = 27 \text{ J}$$

$$W_C = \frac{1}{2} \times 4 \times 6^2 = 72 \text{ J}$$

Series and Parallel Inductors



(a)



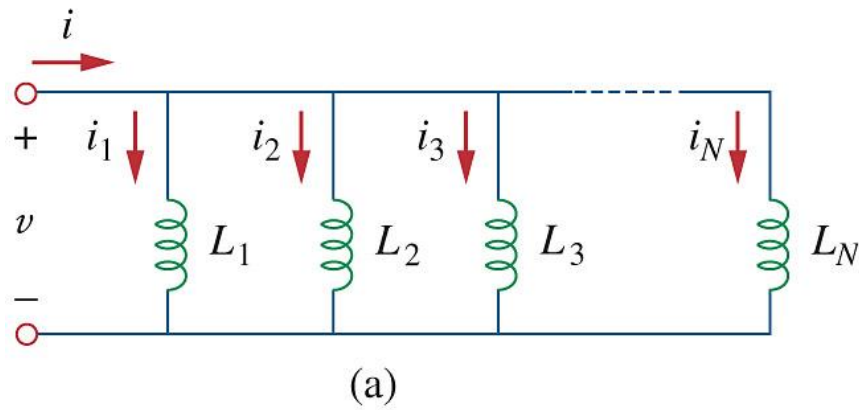
(b)

$$L_{eq} = L_1 + L_2 + \dots + L_N$$

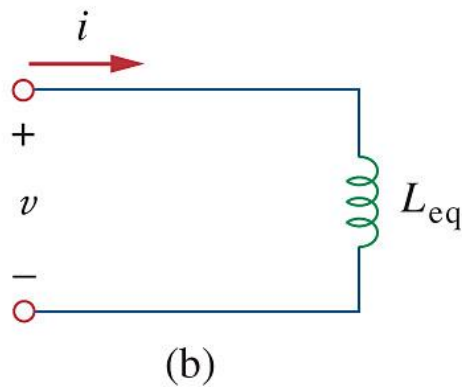
$$V = V_1 + V_2 + \dots + V_N$$

$$L \frac{di}{dt} = L_1 \frac{di}{dt} + \dots + L_N \frac{di}{dt}$$

$$L = L_1 + L_2 + \dots + L_N.$$



$$i = i_1 + i_2 + \dots + i_N.$$



$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N}$$

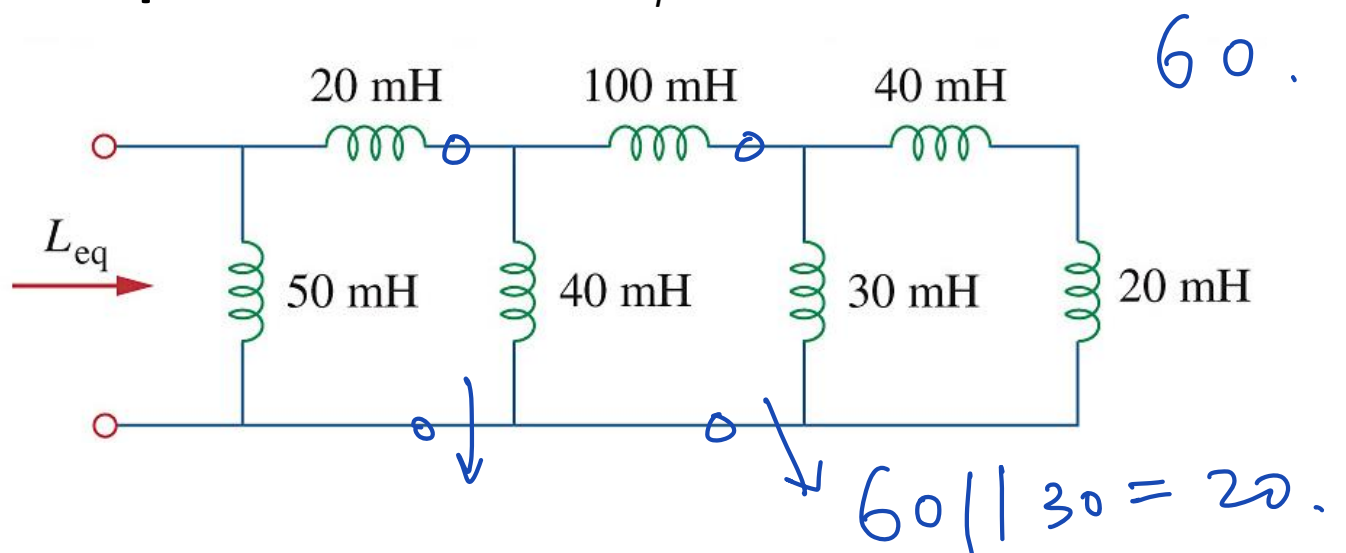
$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \dots + \frac{1}{L_N}.$$

$$L_1 \parallel L_2$$

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2}$$

$$L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$$

Example #4: Calculate L_{eq} .



$$120 || 40 = 30 \text{ mH.}$$

$$L_{eq} = 50 || 50 = 25 \text{ mH.}$$